

Project Abstract: IMDB Natural Language Processing

This project explores the comprehensive application of machine learning, deep learning, and unsupervised methodologies on the benchmark Large Movie Review Dataset (IMDB). The dataset comprises 100,000 highly polar movie reviews, strategically divided into binary classification (50,000 labeled reviews), continuous regression (extracted 1-10 rating scores), and unsupervised subsets (50,000 unlabelled reviews).

The primary objective is to develop end-to-end models capable of extracting underlying semantic sentiment and predicting precise user ratings on a continuous scale. To handle the inherently noisy and unstructured nature of web-scraped natural language, a highly robust data extraction and preprocessing pipeline was engineered. This pipeline transforms raw HTML-injected text files into dense, numerical matrices and tokenized sequences.

The project tracks the evolution of Natural Language Processing techniques by comparing baseline statistical models (Multinomial Naive Bayes, TF-IDF Logistic Regression) against state-of-the-art Deep Learning models (Bidirectional LSTMs, Transformer-based BERT models). Furthermore, we utilized Truncated Singular Value Decomposition (SVD) for dimensionality reduction and lexical VADER analysis to map natural sentiment in the wild.